

NUMERICAL ANALYSIS

PROBLEM SET 3

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We begin by noting that all of the files referenced in this document are available at:

<https://github.com/chtompki/MathSpring2026/tree/main/NumericalAnalysis/ProblemSet3>

1. PROBLEM 1

We'll be approximating $\sqrt{2}$ though this problem

1.1. **1.a.** Our goal is to find a polynomial interpolant $p(x)$ using the Lagrange form given that we have three points in the form $x, f(x)$ as $\{(-1, 1/2), (0, 1), (1, 2)\}$.